

Solving More Equations by Factoring

Solve.

① $3x^3 + 21x^2 + 30x = 0$

$3x(x^2 + 7x + 10) = 0$

$3x(x+2)(x+5) = 0$

ABC=0

$3x=0$ | $x+2=0$ | $x+5=0$
 $x=0$ | $x=-2$ | $x=-5$

③ $4y^3 + 4y^2 - 120y = 0$

$4y(y^2 + y - 30) = 0$

$4y(y+6)(y-5) = 0$

$4y=0$ | $y+6=0$ | $y-5=0$
 $y=0$ | $y=-6$ | $y=5$

⑤ $z^8 - 8z^7 + 16z^6 = 0$

$z^6(z^2 - 8z + 16) = 0$

$z^6(z-4)(z-4) = 0$

$z^6=0$ | $z-4=0$ | $z-4=0$
 $z=0$ | $z=4$ | $z=4$

② $0 = 12y^3 - 27y$

$0 = 3y(4y^2 - 9)$

$0 = 3y(2y+3)(2y-3)$

$3y=0$ | $2y+3=0$ | $2y-3=0$
 $y=0$ | $y=-\frac{3}{2}$ | $y=\frac{3}{2}$

④ $0 = 9x^3 - 36x$

$0 = 9x(x^2 - 4)$

$0 = 9x(x-2)(x+2)$

$9x=0$ | $x-2=0$ | $x+2=0$
 $x=0$ | $x=2$ | $x=-2$

⑥ $0 = 50y^4 - 128y^2$

$0 = 2y^2(25y^2 - 64)$

$0 = 2y^2(5y-8)(5y+8)$

$2y^2=0$ | $5y-8=0$ | $5y+8=0$
 $y=0$ | $y=\frac{8}{5}$ | $y=-\frac{8}{5}$

⑦ $6x^4 + 18x^3 + 12x^2 = 0$

$6x^2(x^2 + 3x + 2) = 0$

$6x^2(x+1)(x+2) = 0$

$6x^2=0$ | $x+1=0$ | $x+2=0$
 $x=0$ | $x=-1$ | $x=-2$

⑧ $0 = 16y^4 - 4y^2$

$0 = 4y^2(4y^2 - 1)$

$0 = 4y^2(2y+1)(2y-1)$

$4y^2=0$ | $2y+1=0$ | $2y-1=0$
 $y=0$ | $y=-\frac{1}{2}$ | $y=\frac{1}{2}$

⑨ $-12x^8 - 4x^7 + 8x^6 = 0$

$-4x^6(3x^2 + x - 2) = 0$

$-4x^6(x+1)(3x-2) = 0$

$-4x^6=0$ | $x+1=0$ | $3x-2=0$
 $x=0$ | $x=-1$ | $x=\frac{2}{3}$

⑩ $0 = -2x^2 + 32$

$0 = -2(x^2 - 16)$

$0 = -2(x-4)(x+4)$

$x-4=0$ | $x+4=0$
 $x=4$ | $x=-4$

⑪ $-6z^3 + 38z^2 - 60z = 0$

$-2z(3z^2 - 19z + 30) = 0$

$-2z(3z-10)(z-3) = 0$

$-2z=0$ | $3z-10=0$ | $z-3=0$
 $z=0$ | $z=\frac{10}{3}$ | $z=3$

⑫ $-5z^4 + 20z^2 = 0$

$-5z^2(z^2 - 4) = 0$

$-5z^2(z-2)(z+2) = 0$

$-5z^2=0$ | $z-2=0$ | $z+2=0$
 $z=0$ | $z=2$ | $z=-2$

$$(13) -18x^6 + 18x^5 + 8x^4 = 0$$

$$-2x^4(9x^2 - 9x - 4) = 0$$

$$-2x^4(3x-4)(3x+1) = 0$$

$$\begin{array}{c|c|c} -2x^4=0 & 3x-4=0 & 3x+1=0 \\ \hline \boxed{x=0} & \boxed{x=\frac{4}{3}} & \boxed{x=-\frac{1}{3}} \end{array}$$

$$(15) 5y^4 + 10y^3 - 40y^2 = 0$$

$$5y^2(y^2 + 2y - 8) = 0$$

$$5y^2(y-2)(y+4) = 0$$

$$\begin{array}{c|c|c} 5y^2=0 & y-2=0 & y+4=0 \\ \hline \boxed{y=0} & \boxed{y=2} & \boxed{y=-4} \end{array}$$

$$(17) 10y^5 - 15y^4 - 25y^3 = 0$$

$$5y^3(2y^2 - 3y - 5) = 0$$

$$5y^3(2y-5)(y+1) = 0$$

$$\begin{array}{c|c|c} 5y^3=0 & 2y-5=0 & y+1=0 \\ \hline \boxed{y=0} & \boxed{y=\frac{5}{2}} & \boxed{y=-1} \end{array}$$

$$(14) 0 = 14y^5 + 49y^4 + 42y^3$$

$$0 = 7y^3(2y^2 + 7y + 6)$$

$$0 = 7y^3(2y+3)(y+2)$$

$$\begin{array}{c|c|c} 7y^3=0 & 2y+3=0 & y+2=0 \\ \hline \boxed{y=0} & \boxed{y=-\frac{3}{2}} & \boxed{y=-2} \end{array}$$

$$(16) 0 = z^4 - 81z^2$$

$$0 = z^2(z^2 - 81)$$

$$0 = z^2(z-9)(z+9)$$

$$\begin{array}{c|c|c} z^2=0 & z-9=0 & z+9=0 \\ \hline \boxed{z=0} & \boxed{z=9} & \boxed{z=-9} \end{array}$$

$$(18) x^3 + 10x^2 - 4x - 40 = 0$$

$$x^2(x+10) - 4(x+10) = 0$$

$$(x+10)(x^2 - 4) = 0$$

$$(x+10)(x-2)(x+2) = 0$$

$$\begin{array}{c|c|c} x+10 & x-2=0 & x+2=0 \\ \hline \boxed{x=-10} & \boxed{x=2} & \boxed{x=-2} \end{array}$$

$$(20) 0 = 4y^3 + 12y^2 - y - 3$$

$$0 = 4y^2(y+3) - 1(y+3)$$

$$0 = (y+3)(4y^2 - 1)$$

$$0 = (y+3)(2y-1)(2y+1)$$

$$\begin{array}{c|c|c} y+3=0 & 2y-1=0 & 2y+1=0 \\ \hline \boxed{y=-3} & \boxed{y=\frac{1}{2}} & \boxed{y=-\frac{1}{2}} \end{array}$$

$$(22) x^3 + 4x^2 + 4x + 16 = 0$$

$$x^2(x+4) + 4(x+4) = 0$$

$$(x+4)(x^2 + 4) = 0$$

$$\begin{array}{c|c} x+4=0 & x^2+4=0 \\ \hline \boxed{x=-4} & x^2 \neq -4 \end{array}$$

$$(19) x^3 - 5x^2 - 9x + 45 = 0$$

$$x^2(x-5) - 9(x-5) = 0$$

$$(x-5)(x^2 - 9) = 0$$

$$(x-5)(x+3)(x-3) = 0$$

$$\begin{array}{c|c|c} x-5=0 & x+3=0 & x-3=0 \\ \hline \boxed{x=5} & \boxed{x=-3} & \boxed{x=3} \end{array}$$

$$(21) 4x^3 + 16x^2 - 9x - 36 = 0$$

$$4x^2(x+4) - 9(x+4) = 0$$

$$(x+4)(4x^2 - 9) = 0$$

$$(x+4)(2x-3)(2x+3) = 0$$

$$\begin{array}{c|c|c} x+4=0 & 2x-3=0 & 2x+3=0 \\ \hline \boxed{x=-4} & \boxed{x=\frac{3}{2}} & \boxed{x=-\frac{3}{2}} \end{array}$$

$$(23) 0 = 5a^3 - 4a^2 + 25a - 28$$

$$0 = a^2(5a-4) + 7(5a-4)$$

$$0 = (5a-4)(a^2 + 7)$$

$$5a-4=0$$

$$\boxed{a = \frac{4}{5}}$$

$$(24) 5z^3 - 35z^2 - z + 7 = 0$$

$$5z^2(z-7) - 1(z-7) = 0$$

$$(z-7)(5z^2-1) = 0$$

$$z-7=0 \quad | \quad 5z^2-1=0$$

$$\boxed{z=7}$$

$$z^2 = \frac{1}{5}$$

$$\sqrt{z^2} = \pm \sqrt{\frac{1}{5}}$$

$$\boxed{z = \pm \frac{\sqrt{5}}{5}}$$

$$(25) 0 = x^3 - 15x^2 - 3x + 45$$

$$0 = x^2(x-15) - 3(x-15)$$

$$0 = (x-15)(x^2-3)$$

$$x-15=0 \quad | \quad x^2-3=0$$

$$\boxed{x=15}$$

$$x^2 = 3$$

$$\sqrt{x^2} = \pm \sqrt{3}$$

$$\boxed{x = \pm \sqrt{3}}$$

$$(26) -2x^5 - 3x^3 + 44x = 0$$

$$-x(2x^4 + 3x^2 - 44) = 0$$

$$-x(2x^2+11)(x^2-4) = 0$$

$$-x(2x^2+11)(x-2)(x+2) = 0$$

$$-x=0 \quad | \quad 2x^2+11=0 \quad | \quad x-2=0 \quad | \quad x+2=0$$

$$\boxed{x=0} \quad | \quad x^2 \neq -\frac{11}{2} \quad | \quad \boxed{x=2} \quad | \quad \boxed{x=-2}$$

$$(27) 0 = 4z^7 + 44z^5 + 112z^3$$

$$0 = 4z^3(z^4 + 11z^2 + 28)$$

$$0 = 4z^3(z^2+7)(z^2+4)$$

$$4z^3=0 \quad | \quad z^2+7=0 \quad | \quad z^2+4=0$$

$$\boxed{z=0} \quad | \quad z^2 \neq -7 \quad | \quad z^2 \neq -4$$

$$(28) 5z^7 - 5z^3 = 0$$

$$5z^3(z^4-1) = 0$$

$$5z^3(z^2-1)(z^2+1) = 0$$

$$5z^3(z-1)(z+1)(z^2+1) = 0$$

$$5z^3=0 \quad | \quad z-1=0 \quad | \quad z+1=0 \quad | \quad z^2+1=0$$

$$\boxed{z=0}$$

$$\boxed{z=1}$$

$$\boxed{z=-1}$$

$$z^2 \neq -1$$

$$\boxed{x=0}$$

$$\boxed{x=2}$$

$$\boxed{x=-2}$$

$$x^2 \neq 4$$

$$(29) 0 = 3x^6 - 48x^2$$

$$0 = 3x^2(x^4-16)$$

$$0 = 3x^2(x^2-4)(x^2+4)$$

$$0 = 3x^2(x-2)(x+2)(x^2+4)$$

$$3x^2=0 \quad | \quad x-2=0 \quad | \quad x+2=0 \quad | \quad x^2+4=0$$

$$(30) 2x^3 + 4x^2 = -2x$$

$$2x^3 + 4x^2 + 2x = 0$$

$$2x(x^2+2x+1) = 0$$

$$2x(x+1)(x+1) = 0$$

$$2x=0 \quad | \quad x+1=0 \quad | \quad x+1=0$$

$$\boxed{x=0} \quad | \quad \boxed{x=-1} \quad | \quad \boxed{x=-1}$$

$$(32) 3x^7 = -8x^6 + 35x^5$$

$$3x^7 + 8x^6 - 35x^5 = 0$$

$$x^5(3x^2+8x-35) = 0$$

$$x^5(3x-7)(x+5) = 0$$

$$x^5=0 \quad | \quad 3x-7=0 \quad | \quad x+5=0$$

$$\boxed{x=0} \quad | \quad \boxed{x=\frac{7}{3}} \quad | \quad \boxed{x=-5}$$

$$(34) x^3 - 2x^2 = x - 2$$

$$x^3 - 2x^2 - x + 2 = 0$$

$$x^2(x-2) - 1(x-2) = 0$$

$$(x-2)(x^2-1) = 0$$

$$(x-2)(x+1)(x-1) = 0$$

$$\boxed{x=2} \quad | \quad \boxed{x=-1} \quad | \quad \boxed{x=1}$$

$$(31) 72x^2 = 8x^4$$

$$0 = 8x^4 - 72x^2$$

$$0 = 8x^2(x^2-9)$$

$$0 = 8x^2(x+3)(x-3)$$

$$8x^2=0 \quad | \quad x+3=0 \quad | \quad x-3=0$$

$$\boxed{x=0} \quad | \quad \boxed{x=-3} \quad | \quad \boxed{x=3}$$

$$(33) 20x^3 = 45x$$

$$0 = -20x^3 + 45x$$

$$0 = -5x(4x^2-9)$$

$$0 = -5x(2x-3)(2x+3)$$

$$-5x=0 \quad | \quad 2x-3=0 \quad | \quad 2x+3=0$$

$$\boxed{x=0} \quad | \quad \boxed{x=\frac{3}{2}} \quad | \quad \boxed{x=-\frac{3}{2}}$$

$$(35) -9y + 15 = 6y^3 - 10y^2$$

$$0 = 6y^3 - 10y^2 + 9y - 15$$

$$0 = 2y^2(3y-5) + 3(3y-5)$$

$$0 = (3y-5)(2y^2+3)$$

$$3y-5=0 \quad | \quad 2y^2+3=0$$

$$\boxed{y=\frac{5}{3}} \quad | \quad y^2 \neq -\frac{3}{2}$$

• 36) $x^2 + 5x = 0$

$$x(x+5) = 0$$

$$\boxed{x=0} \quad | \quad \boxed{x+5=0}$$

$$\boxed{x=-5}$$

38) $3x^2 + 15x^3 = 0$

$$3x^2(1+5x) = 0$$

$$\boxed{3x^2=0} \quad | \quad \boxed{1+5x=0}$$

$$\boxed{x=0} \quad | \quad \boxed{x=-\frac{1}{5}}$$

40) $0 = -8x - 2x^2$

$$0 = -2x(4+x)$$

$$\boxed{-2x=0} \quad | \quad \boxed{4+x=0}$$

$$\boxed{x=0} \quad | \quad \boxed{x=-4}$$

37) $18x^2 = 42x$

$$18x^2 - 42x = 0$$

$$6x(3x-7) = 0$$

$$\boxed{6x=0} \quad | \quad \boxed{3x-7=0}$$

$$\boxed{x=0} \quad | \quad \boxed{x=\frac{7}{3}}$$

39) $20x = 10x^2$

$$0 = 10x^2 - 20x$$

$$0 = 10x(x-2)$$

$$\boxed{10x=0} \quad | \quad \boxed{x-2=0}$$

$$\boxed{x=0} \quad | \quad \boxed{x=2}$$

41) $-11x^3 = 44x^2$

$$0 = 11x^3 + 44x^2$$

$$0 = 11x^2(x+4)$$

$$\boxed{11x^2=0} \quad | \quad \boxed{x+4=0}$$

$$\boxed{x=0} \quad | \quad \boxed{x=-4}$$